

Standard NARWC Survey Format

Reference: `src/+narwc/+io/+parsers/StandardFormat.m`.

Overview

Comma-delimited CSV with 55 columns in a fixed legacy order (no header row naming the columns — `StandardFormat` supplies the names itself from `CSV_FIELD_ORDER`). This is the format of the historical monolithic legacy CSV being migrated (Phase 1).

File Format

- Delimiter: Comma (,)
- Header Row: None. Column names are not read from the file — they come from `StandardFormat.CSV_FIELD_ORDER`. Note: the parser's import options set `DataLines = [2, Inf]`, so the first line of the file is always skipped regardless of content.
- Missing Values: NULL, ., or empty string
- Extra columns beyond the 55 are ignored (`ExtraColumnsRule = 'ignore'`)

Field Definitions

Column order and types below are read directly from `StandardFormat.CSV_FIELD_ORDER` and `src/+narwc/+db/FieldDefinitions.m` (the canonical source of truth for all 55 field names/types) — not hand-maintained, so re-generate this table from those two files if either changes.

Position	Field Name	Type	Description
1	ALT	double	Altitude in meters
2	ANHEAD	double	Angle to head
3	BEAUFORT	double	Beaufort sea state (0-9)
4	BEHAV1	double	Behavior code 1
5	BEHAV2	double	Behavior code 2
6	BEHAV3	double	Behavior code 3
7	BEHAV4	double	Behavior code 4
8	BEHAV5	double	Behavior code 5
9	BEHAV6	double	Behavior code 6
10	BEHAV7	double	Behavior code 7
11	BEHAV8	double	Behavior code 8
12	BEHAV9	double	Behavior code 9
13	BEHAV10	double	Behavior code 10
14	BEHAV11	double	Behavior code 11

Position	Field Name	Type	Description
15	BEHAV12	double	Behavior code 12
16	BEHAV13	double	Behavior code 13
17	BEHAV14	double	Behavior code 14
18	BEHAV15	double	Behavior code 15
19	BLOCK	string	Survey block identifier
20	CLOUD	double	Cloud cover (0-10)
21	CONFIDNC	double	Confidence level
22	DAY	double	Day of month (1-31)
23	DDSOURCE	string	Data source code
24	EVENTNO	double	Event number
25	FILEID	string	File/Survey identifier
26	GLAREL	double	Glare level left
27	GLARER	double	Glare level right
28	HEADING	double	Ship/aircraft heading (degrees)
29	IDREL	double	ID reliability
30	IDSOURCE	string	ID source
31	LAT_DD	double	Latitude (decimal degrees)
32	LEGNO	double	Leg number
33	LEGSTAGE	double	Leg stage
34	LEGTTYPE	double	Leg type
35	LONG_DD	double	Longitude (decimal degrees)
36	MONTH	double	Month (1-12)
37	NUMBER	double	Number of animals
38	NUMCALF	double	Number of calves
39	PHOTOS	double	Number of photos
40	PLATFORM	double	Platform code
41	S_LAT	double	Starting latitude
42	S_LONG	double	Starting longitude
43	S_TIME	double	Starting time
44	SIGHTNO	double	Sighting number
45	SPECCODE	string	Species code
46	STRATUM	string	Survey stratum
47	STRIP	double	Strip number
48	SURFTEMP	double	Surface temperature
49	TAXCODE	double	Taxonomic code
50	TIME	double	Time of observation (HHMMSS)
51	VISIBLTY	double	Visibility
52	WX	string	Weather code
53	YEAR	double	Year
54	ANGLEL	double	Angle left of trackline
55	ANGLER	double	Angle right of trackline

Example

```
244,NULL,4,...  
244,NULL,4,...
```

(55 comma-separated values per row, in the position order above, no header line — remember the parser always skips physical line 1.)

Validation Notes

These are the actual configured defaults (`config/defaults/validation_config_default.m`) as of this writing — always verify against that file directly, since batch configs (e.g. `migration.m`) can override them per run:

- LAT_DD/LONG_DD hard bounds (blocking error): ± 90 / ± 180 (physical validity only).
- LAT_DD/LONG_DD survey-area bounds (warning, not error): lat 20–55, lon -85 to -40 (`coordinate_rules.outside_survey_lat/outside_survey_lon`).
- YEAR hard bounds (blocking error): 1900 to current year + 1.
- YEAR warning threshold: below 1980 (`datetime_rules.year_too_old`).
- SPECCODE must match `data/tables/SPECCODE.csv` (see `species_rules.m`).